Calculus I Assignment 7

1. Over which intervals are the following functions convex?

   (a) $f(x) = 5x^4 - 3x^3 + x^2 - 1$

   (b) $f(x) = \frac{x+1}{x-1}$

   (c) $f(x) = \sqrt{x}$

   (d) $f(x) = \frac{1}{\sqrt{x}}$

2. Find an interval $(0, b)$ where $\mathrm{e}^{-1/x}$ is convex. Sketch the graph on $(0, \infty)$.

3. Find the limit. Use l'Hospital's Rule where appropriate. If there is a more elementary method, consider using it. If l'Hospital's Rule doesn't apply, explain why.

   (a) $\lim_{x\to\pi/4} \frac{\sin x - \cos x}{\tan x - 1}$

   (b) $\lim_{x\to 0} \frac{\sqrt{1+2x} - \sqrt{1-4x}}{x}$

   (c) $\lim_{x\to 1} \frac{x \sin(x-1)}{2x^2 - x - 1}$

   (d) $\lim_{x\to 0^+} x^{\sqrt{x}}$

   (e) $\lim_{x\to 0^+} (1 - \cos x)^{\sin x}$

   (f) $\lim_{x\to\infty} \left(\frac{2x-3}{2x+5}\right)^{2x+1}$

4. Evaluate
$$\lim_{x\to\infty} \left[x - x^2 \ln\left(\frac{1+x}{x}\right)\right]$$

5. For what values of a and b is the following equation true?
$$\lim_{x\to 0} \left(\frac{\sin 2x}{x^3} + a + \frac{b}{x^2}\right) = 0$$

6. Verify by differentiation that the formula is correct.

   (a) $\int \ln x dx = x \ln x - x + C$

   (b) $\int x\sqrt{a+bx} dx = \frac{2}{15b^2}(3bx - 2a)(a+bx)^{3/2} + C$

7. Find the general indefinite integral.

   (a) $\int \sqrt{t}\,(t^2 + 3t + 2)\, dt$

   (b) $\int \left(x^2 + 1 + \frac{1}{x^2+1}\right) dx$

   (c) $\int \left(\frac{1+r}{r}\right)^2 dr$

   (d) $\int (2 + \tan^2\theta)\, d\theta$

8. Evaluate the integral by making the given substitution.

(a) $\int x^2\sqrt{x^3+1}dx, \quad u = x^3+1$

(b) $\int \sin^2\theta\cos\theta d\theta, \quad u = \sin\theta$

(c) $\int \frac{1}{x^2}\sqrt{1+\frac{1}{x}}dx, \quad u = 1+\frac{1}{x}$

9. Evaluate the indefinite integral.

(a) $\int x\sqrt{1-x^2}dx$

(b) $\int (5-3x)^{10}dx$

(c) $\int t^3 e^{-t^4}dt$

(d) $\int \sin t\sqrt{1+\cos t}dt$

(e) $\int \frac{e^u}{(1-e^u)^2}du$

(f) $\int \frac{a+bx^2}{\sqrt{3ax+bx^3}}dx$

(g) $\int \frac{(\ln x)^2}{x}dx$

(h) $\int \sin x \sin(\cos x)dx$

Bonus Questions

1. Is e^f convex when f is convex?

2. Give an example of convex functions f and g for which $f \circ g$ is not convex.